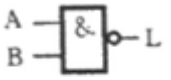


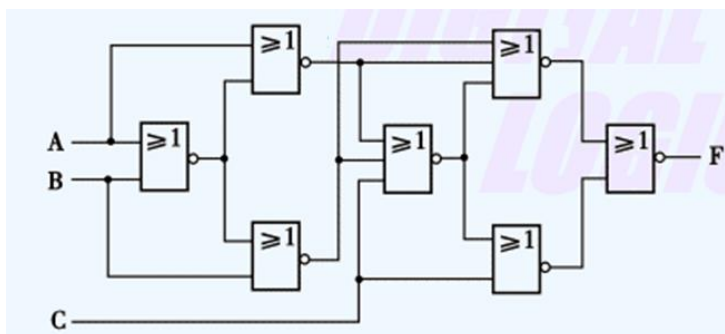
Assignment of Chapter Three (A)



A. Fill-in-Blank:

- 1 The universal gates usually refer to NAND and NOR, such two logic gates, their logic circuit symbols are: and .
- 2 A combinational logic circuit stable output signals at any time, replies on its input signals at this moment.
- 3 The relationships between the input and output of a combinational logic circuit can be represented by its truth table/ logic function expression.
- 4 "The NAND, NOR, AND-OR-INVERT logic gates can implement the basic three logic funtions." This statement is true (true/false).
- 5 "The one input of an X-OR gate is always 0, such X-OR can be used as NOT gate." This statement is false (true/false).
- 6 "Any logic circuit containing the basic logic gates, can be called a combinational logic circuit." This statement is false (true/false).

B. Analyze the following logic circuit:



$$F = A \oplus B \oplus C = A \oplus B \oplus \bar{C}$$

When the input of ABC are 000, 011, 101, 110, the output F is 1; otherwise, the output is 0.

